

Input Devices

text input, positional input, (and gestural)

Input Devices

- Keyboard
- Pointing devices
 - Mouse
 - Trackball
 - Joystick
 - Pen
 - Touch screen
- Speech
- Video/Image/Gesture
 - Camera
 - Sensors





The wheel. Reinvented.

MacBook Wheel

One button. Endless possibilities.

MacBook Wheel (The Onion)

- <https://youtu.be/9BnLbv6QYcA>

iPod Wheel

- It works great here



Specific vs. General Input Devices

- General input devices adapted to many task
 - Problems?
- Specific input devices optimized for specific tasks
 - Problems?



Dimensions to Classify Computer Input

- Sensing Method
 - mechanical (e.g. switch)
 - motion (e.g. accelerometer, gyroscope)
 - contact (e.g. capacitive touch, pressure sensor)
 - signal processing (e.g. computer vision, audio)
- Continuous vs. Discrete
- Degrees of Freedom (DOF)
 - number of independent motions that are allowed to the body



Dimensions to Classify Computer Input

- iPod wheel: 1 DOF contact sensing continuous, 1 DOF discrete mechanical buttons (assume one button at a time)
- Mouse: 2 DOF computer vision (used to be mechanical), 1 DOF mechanical wheel, 1 DOF discrete mechanical buttons
- Wii: 6 DOF continuous translation and rotation, 1 DOF discrete mechanical buttons
- keyboard: ~1 DOF discrete mechanical keys



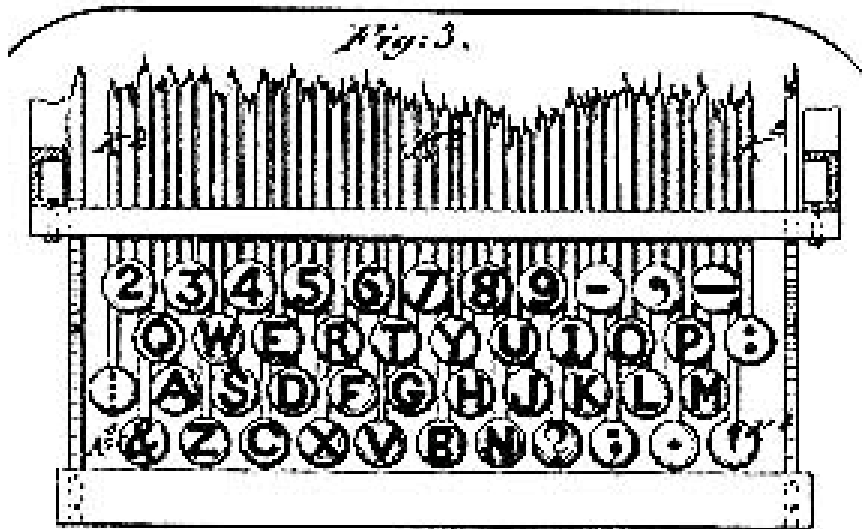
The keyboard



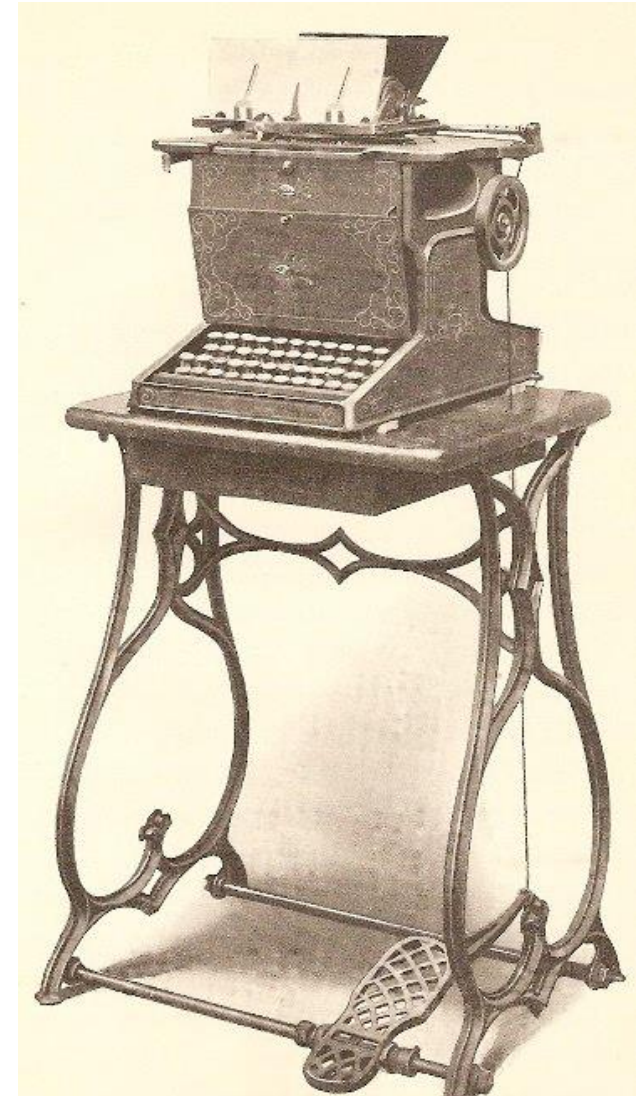
Clavichords in 14th century
Musical keyboard pre-date typewriters by ~500 years

Typewriters and QWERTY

- Christopher Latham Sholes
 - [invented the QWERTY keyboard](#)
- QWERTY not designed to slow typing down
→ [designed to space "typebars" to reduce jams and speed typing up](#)



1874 QWERTY patent drawing



Remington

<https://www.youtube.com/watch?v=WEyCINKkR-Q>

Perceived Qwerty Problems

- Common combinations require awkward finger motions. (eg: tr)
- Common combinations require a jump over home row. (eg: br)
- Common combinations are typed with one hand. (e.g. was, were)
- Most typing with the left hand
- About 16% of typing uses lower row, 52% top row, 32% home row



DVORAK (<https://dvorak-keyboard.com/>)

- Alternative layout for two-handed keyboard

Dvorak



Qwerty



Dvorak Typing Movement Optimizations

- Letters should be typed by alternating between hands
- For maximum speed and efficiency, the most common letters and digraphs should be the easiest to type. Thus, about 70% of keyboard strokes are on home row.
- The least common letters should be on the bottom row, which is the hardest row to reach.
- The right hand should do more of the typing, because most people are right-handed.



Qwerty vs. Dvorak

- In studies, sometimes DVORAK a bit faster, sometimes QWERTY a bit faster
- This implies *no significant difference*
- The Great Keyboard Debate: QWERTY versus Dvorak (Link [here](#))



August Dvorak

Mechanical Design of Keyboards

- To increase portability of devices, keyboards are frequently downsized: low-profile keys, smaller keys
- Many shapes, foldable, roll-up
- All interfere with typing
- Much more significant problem than Dvorak vs Qwerty keyboards

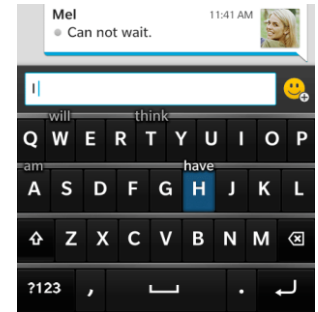


“Virtual” Keyboards

- Touch screen or other flat surface
- Problems:
 - small keys reduce accuracy
 - no mechanical feedback makes it hard to tell if key was pressed
 - no tactile feedback makes it hard to find the home row
 - resting of hands difficult
- Advantage:
 - portable, no extra hardware
 - customizable keys (e.g. new language, symbols, emojis)
 - customizable layout or functionality (e.g. swipe, thumb layout)



iPhone



z10



Virtual projection keyboard
https://www.youtube.com/watch?v=FdnOED8_66s



Tap Strap 2

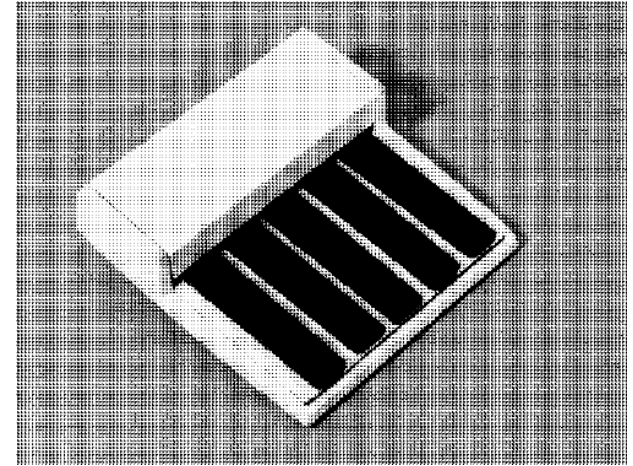


TapXR

<https://www.tapwithus.com/>

Chording Keyboards

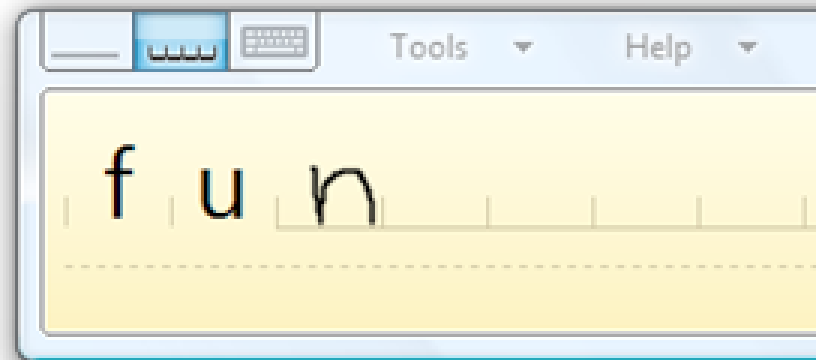
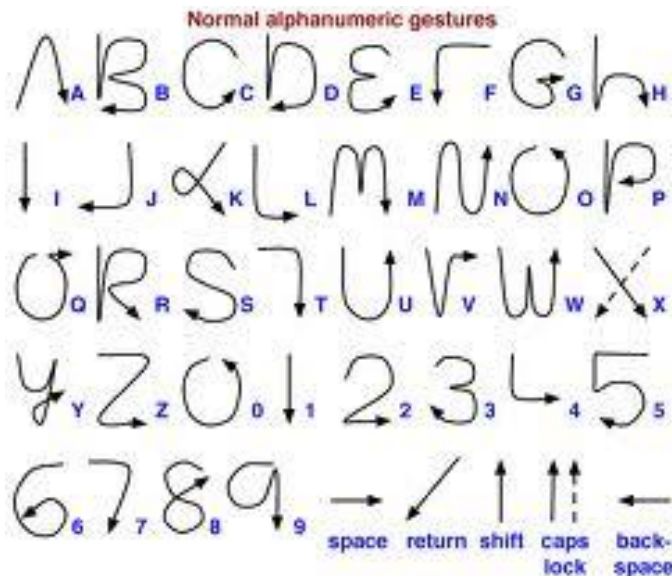
- Englebart's NLS Keyboard
 - Multiple keys together produce letter
 - No "targeting", potentially very fast
 - Can be small and portable
 - One handed
- [Thad Starner](#)'s Twiddler
 - for wearable computing input



<https://twiddler.tekgear.com/>

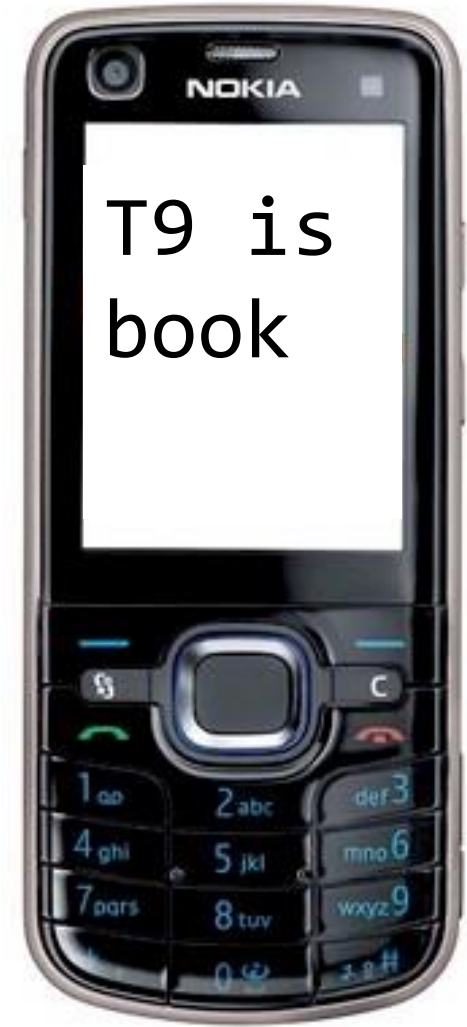
Text Recognition and Gestures

- Graffiti/Unistroke Gestures
 - map single strokes to “enter letter” commands
- Natural Handwriting recognition
 - dictionary-based classification algorithms



Predictive Text

- Use language characteristics to predict input
 - Given characters typed so far, what are the most likely intended words?
 - Given words typed so far, what is the most likely word to follow?
- A variation is used for T9, nine-key text entry
 - Given an ambiguous set of characters, what is the most likely word
- Possible Problems
 - “collisions” between common words
 - entering words not in dictionary difficult
 - hard to focus on typing and monitoring prediction



Other Predictive Text Systems

- **Apple QuickType**

- Apple's predictive text feature that enables the software to build custom dictionaries.

- **Google Gboard**

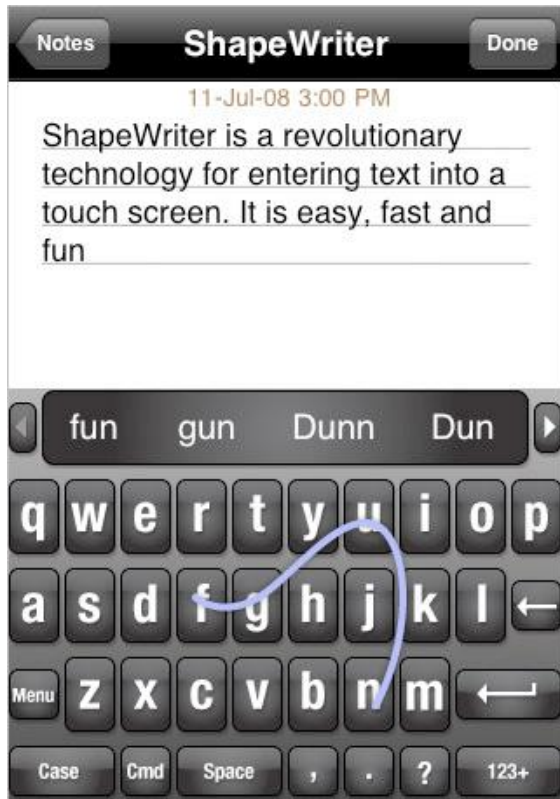
- Gboard is sometimes also called the predictive text suggestion strip, word suggestion or auto-suggest

- **Typewise**

- compatible with both the Apple iOS and Android operating systems

- Predictive text uses machine learning to guess a writer's next word based on their writing style.
- As the user types, the machine learning algorithm notes commonly used words and creates a personalized dictionary.

Gestural Text Input



ShapeWriter

<https://www.shuminzhai.com/shapewriter>



8Pen Keyboard

<http://www.8pen.com/>

Text Input Expert-User Input Rates

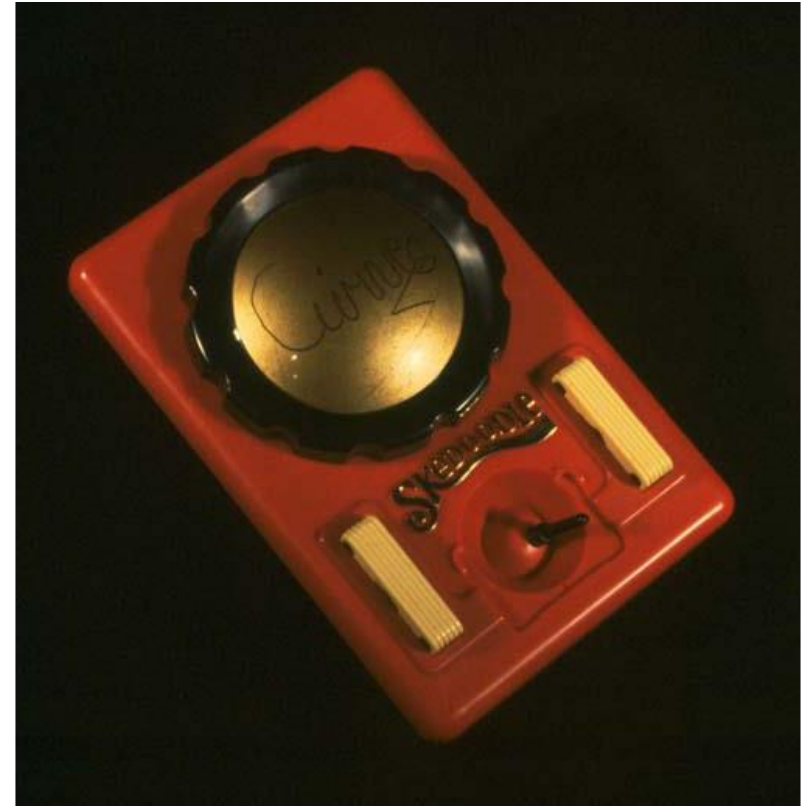
Device	Input Rates
Qwerty Desktop	80+ WPM typical, record: 150 WPM for 50 minutes
Qwerty Thumb	60 WPM typical with training (Clarkson et al., CHI 2005)
Soft Keyboards	45 WPM
T9	45 WPM possible for experts (Silverberg et al., CHI 2000)
Gestural	~30 WPM 8Pen, ShapeWriter claims 80 WPM (expert)
Handwriting	33 WPM (Wilkund et al., Human Factors Society, 1987)
Graffiti 2	9 WPM (Koltringer, Grechenig, CHI 2004)

Positional Input



Etch-A-Sketch

<http://youtu.be/hq3Et9gOISI>



Skedoodle

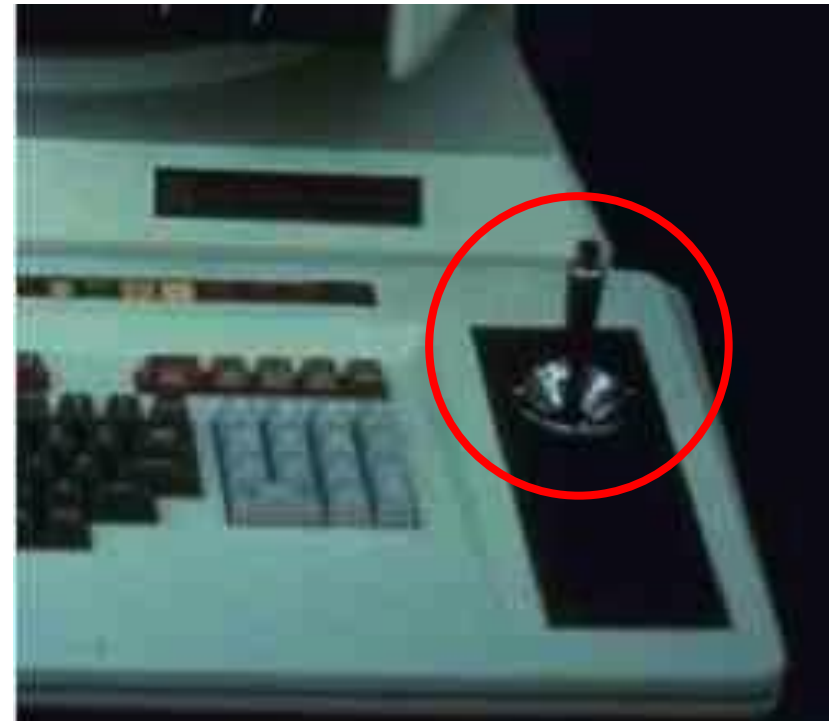
<http://youtu.be/ic1rbFGhJ8g>

Images from <http://research.microsoft.com/en-us/um/people/bibuxton/buxtoncollection/browse.aspx>



<http://www.pdp8.net/tek4010/tek4010.shtml>

Tektronix 4010



From Buxton, et al, *Human Input to Computer Systems: Theories, Techniques, and Technologies*

Absolute vs. Relative Position

- Absolute position is a direct mapping (i.e., 1-to-1) of input device position to a display input position



Indirect



Direct

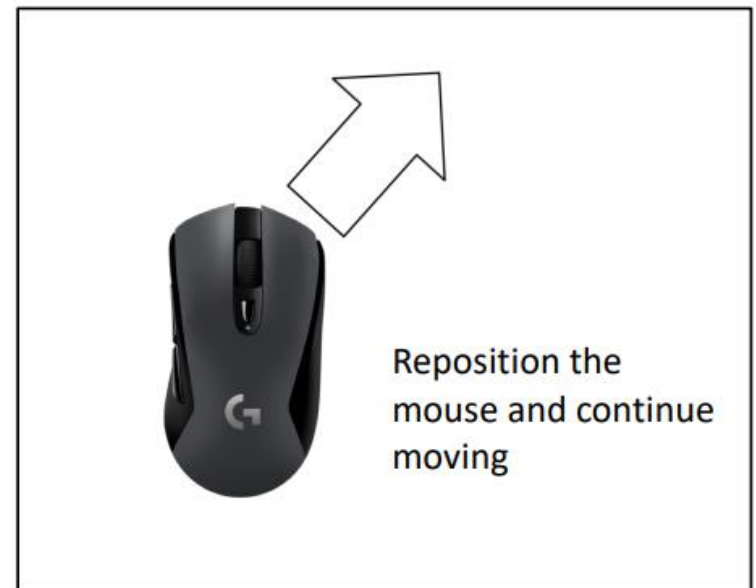
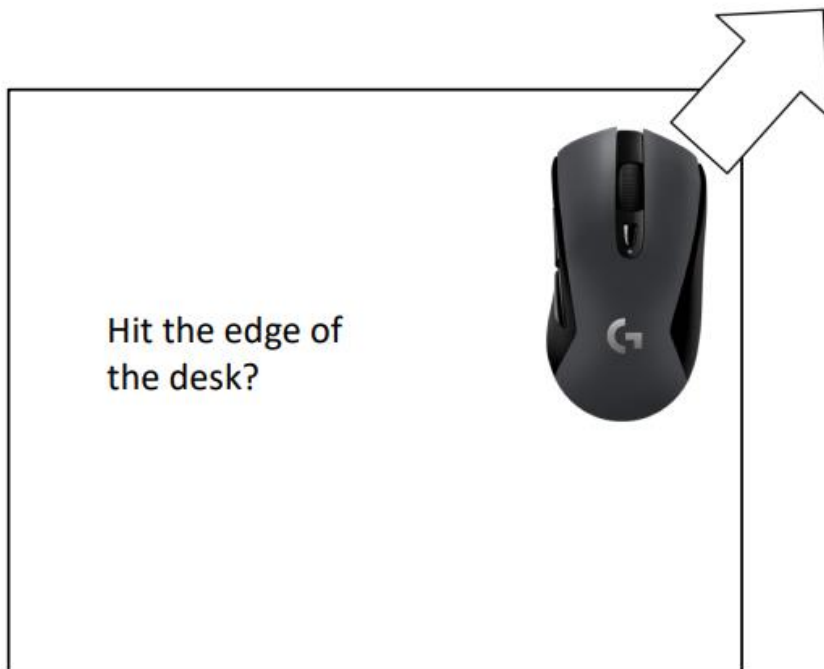
- Relative position maps changes in input device position to changes in display input position



- To make relative position work, you need a “clutch”

Clutching

- **Scenario:** you're moving the mouse and you hit the edge of the desk before you finish positioning the mouse. What do you do?
- To make relative position work, you need to clutch (i.e. repeatedly move to achieve the target)
 - clutching is one solution to making relative motion work



Direct vs. Indirect Input

- Indirect:
 - The position of the cursor is controlled by some external devices

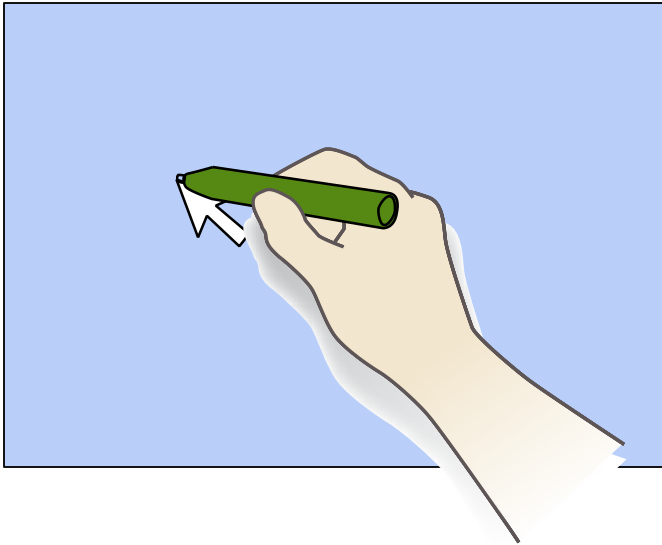


- Direct:
 - the position of the cursor is controlled by direct contact with the screen.

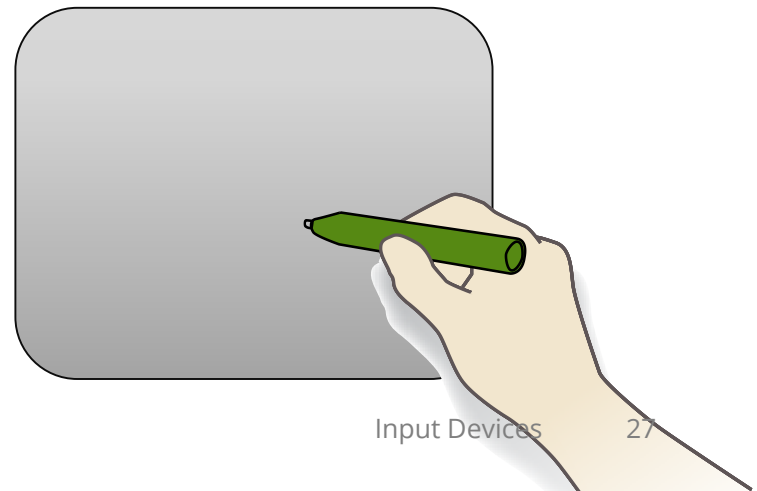
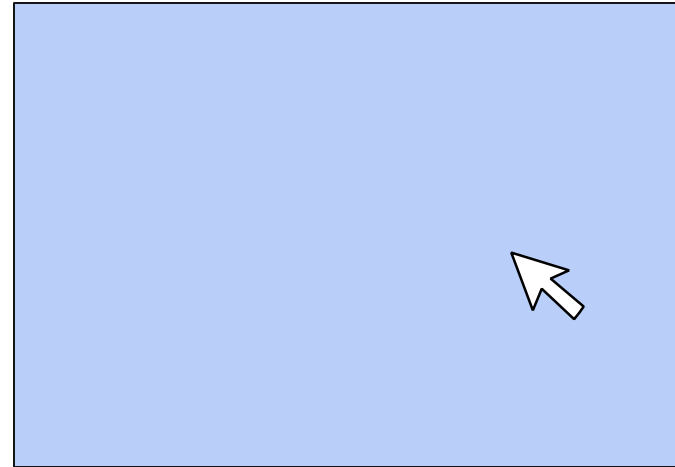


Combination

Absolute Direct



Absolute Indirect

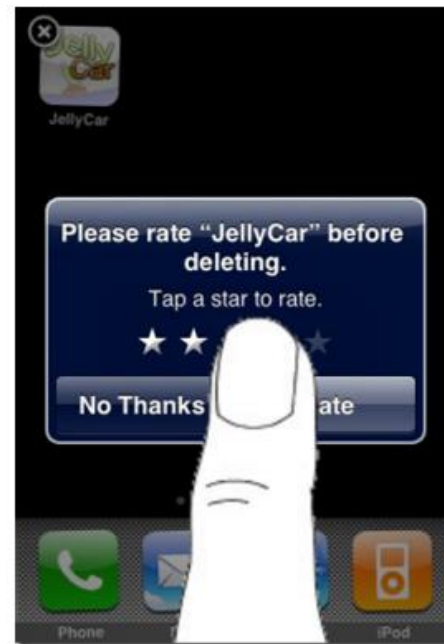


Direct input

- Occlusion:
 - The finger covers the object of interest.



Here, the letter under the finger grows and moves upwards to reduce the problem.



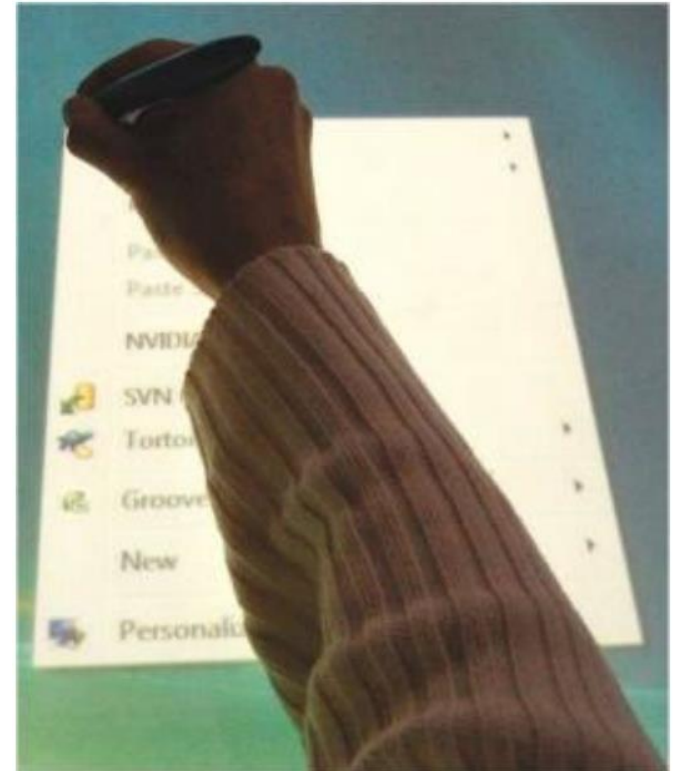
Problematic design



Better design

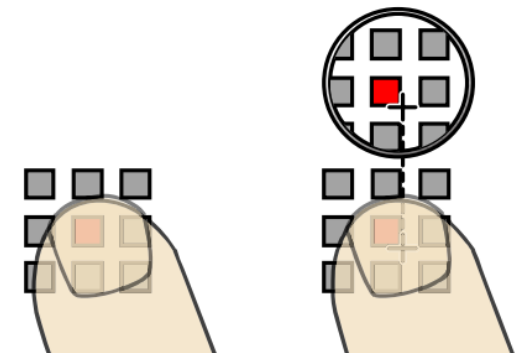
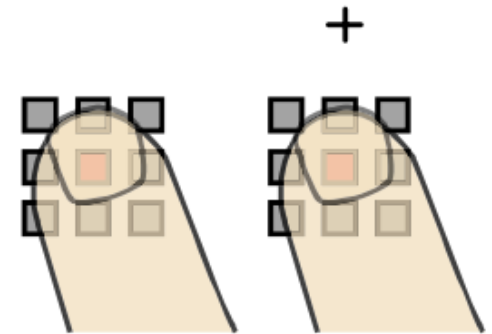
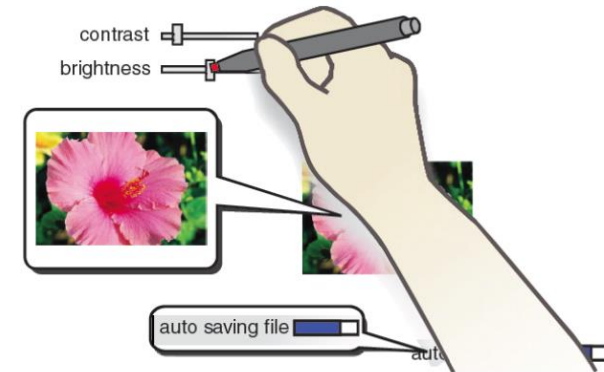
Direct input

- User's hand, fingers or forearm **cover large portions of the display**
- This phenomena referred to as **occlusion** – which creates problems not experienced with conventional mouse input
- Occlusion likely contributes to errors, increases fatigue, forces inefficient movements, and impedes performance



Occlusion-Aware Interfaces

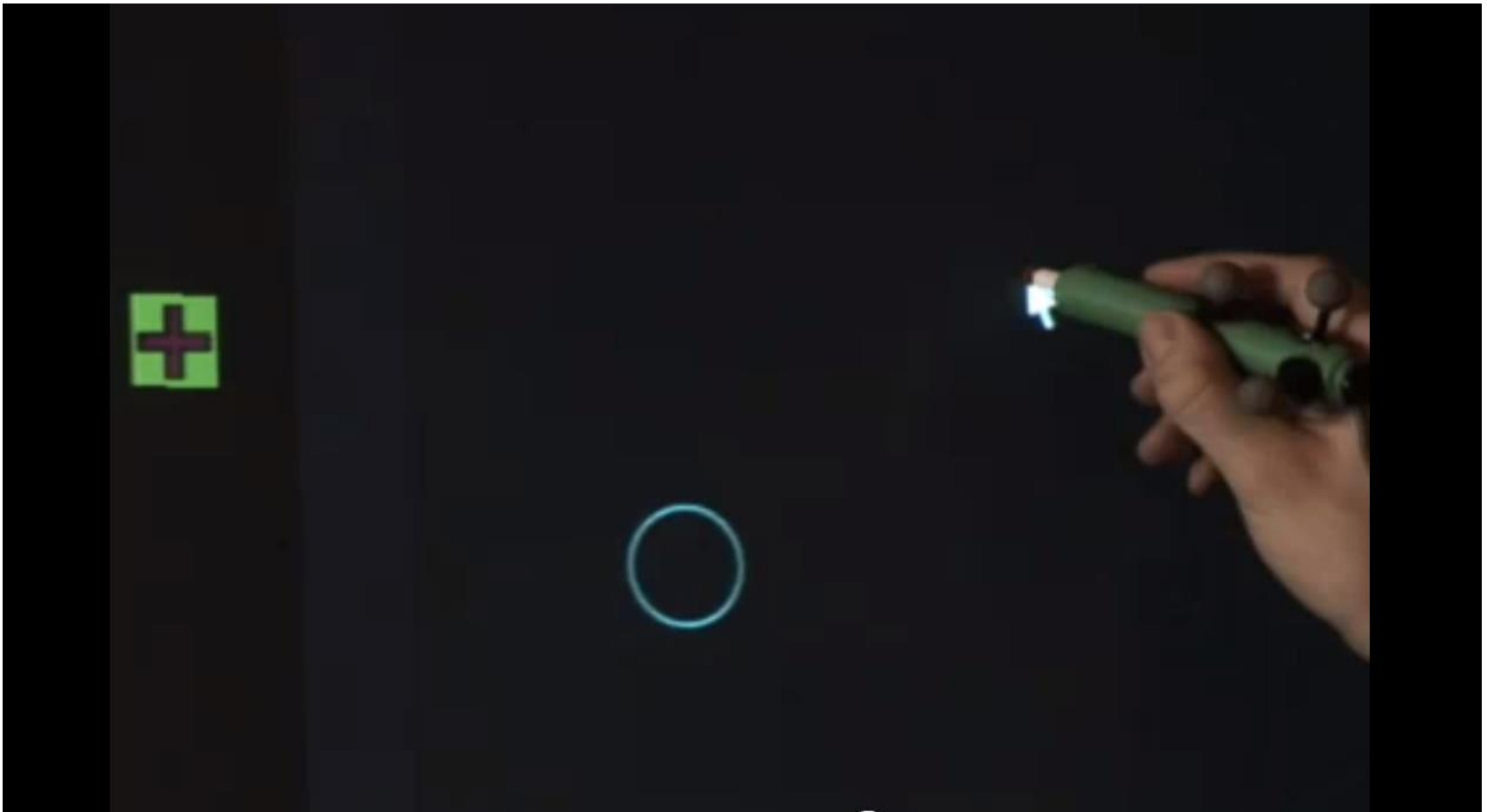
- Occlusion-Aware Viewer technique: [Vogel & Balakrishnan](#), 2010
 - Displays otherwise missed previews and status messages in a non-occluded area using a bubble-like callout.
- Offset Cursor. B. Shneiderman, 1991
 - Create an offset between pointer and finger
 - Cannot aim for the actual target anymore
 - Unreachable areas (e.g., bottom of the screen)
- Shift. [Vogel and Baudisch](#). 2007.
 - Shift creates a callout showing a copy of the occluded screen area and places it in a non-occluded location



Hybrid Absolute and Relative Pointing:

Technique to let users switch between absolute and relative pointing with a direct input device such as a pen

- <http://youtu.be/FZmOBlg5KjM>

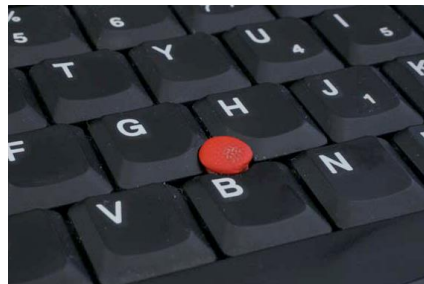


Relative Pointing: Mappings

- Position control: maps human input to the position of the cursor (or object of interest)
 - Examples: mouse, touchpad



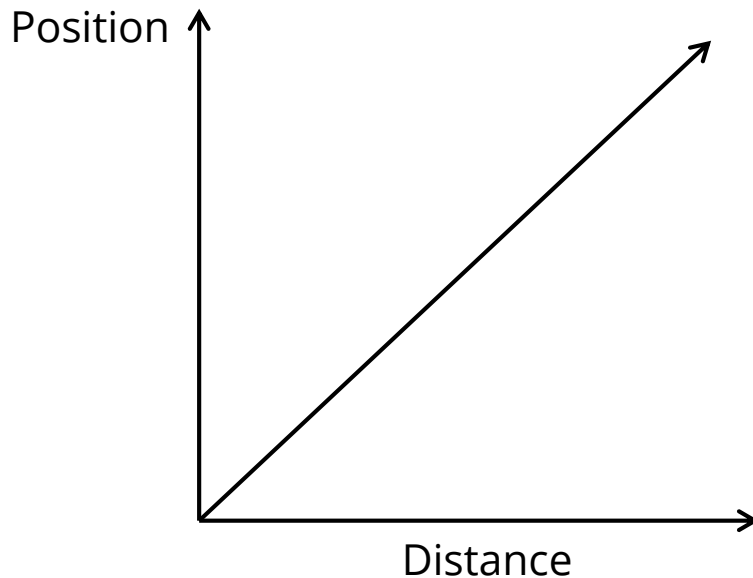
- Rate (or velocity) control: maps human input to the velocity of the cursor (or object of interest)
 - Examples: joystick, trackpoint



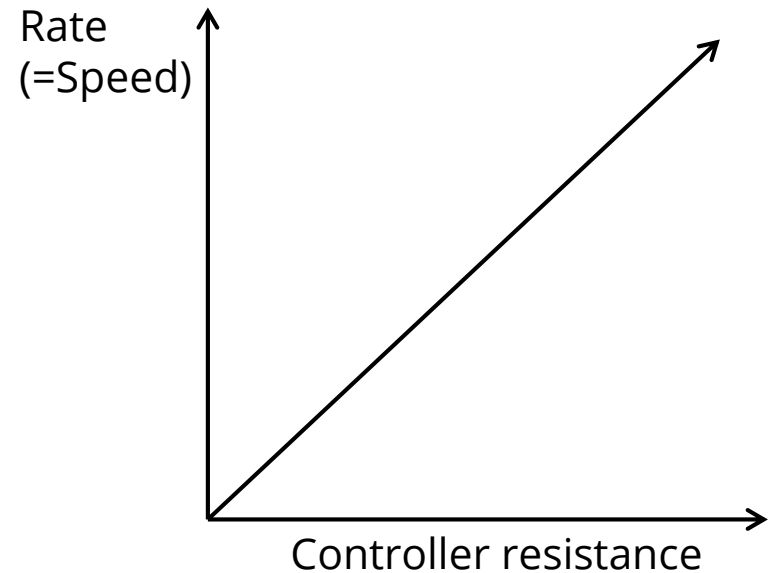
Position vs. Rate Control Transfer Function

- Displacement sensing should be mapped to position
- Force sensing should be mapped to rate

Displacement sensing
(Position control)



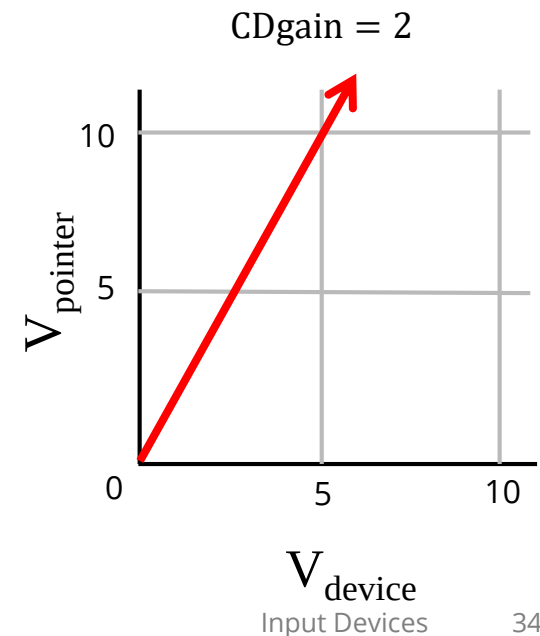
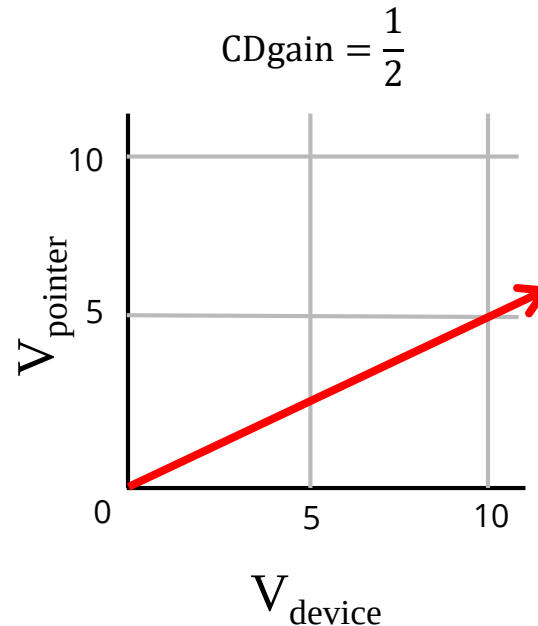
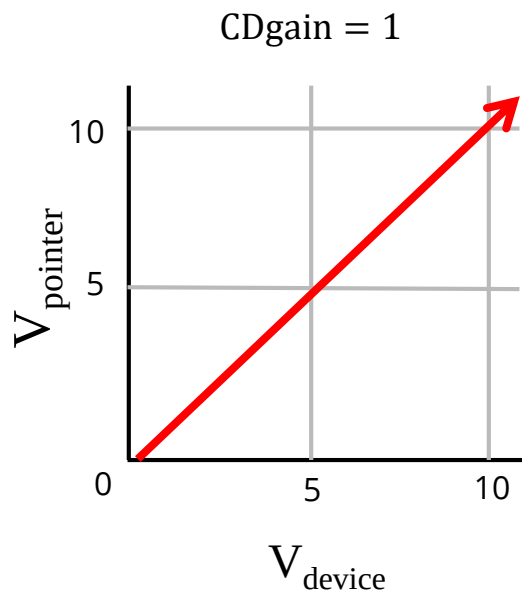
Force sensing
(rate control)



Control-Display Gain (CD Gain)

- Maps the movement of the pointing device to the movement of the display pointer
 - Ratio of **display pointer** movement to **device control** movement

$$\text{CDgain} = V_{\text{pointer}} / V_{\text{device}}$$



Properties of Positional Input Devices

- Absolute vs. Relative Positioning
 - touchscreen = absolute
 - mouse = relative
- Direct vs. Indirect Contact
 - direct = touchscreen
 - indirect = mouse
- Force vs. Displacement Sensing
 - (most) joysticks = force
 - mouse = displacement
- Position vs. Rate Control
 - (most) joysticks = rate
 - mouse = position
- DOF (Dimensions) Sensed
 - 2 = mouse, 6 = Wiimote



Speech Input

- Available in commercial applications and devices (e.g., smartphones)
- <https://www.youtube.com/watch?v=f4dIAV4E2UA>
- Accuracy in discrete word recognition is higher than that in continuous speech recognition
- With AI, speech recognition system is now almost as accurate as humans
- Voice-activated personal assistant platforms such as Alexa and Siri continue to grow in popularity

Speech Input

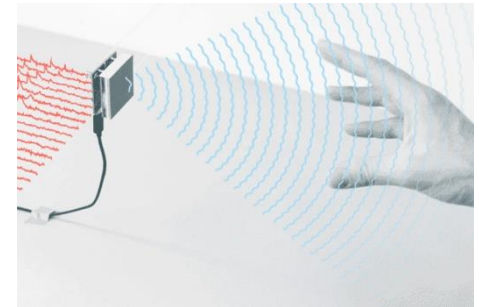
- Advantages:
 - Much training is not needed
 - Good for the case when hands/eyes are occupied
- Disadvantages:
 - Feedback may be limited (Verification procedures slow down speed)
 - Speech is a single channel mode: cannot speak while listening, cannot listen to >1 messages simultaneously
 - Not appropriate for use in open or noisy environments
- Guidelines
 - Consider voice recognition as an input device when user's hands are occupied or user's eyes are busy
 - Avoid voice recognition as an input device in open environments (noise & security)

Gestural Input

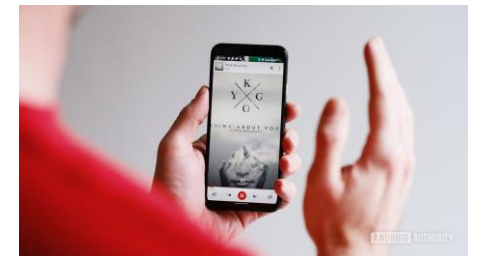
- Gestures map movements to commands
 - instead of pressing button, “wave your hand”
- Leap Motion controller
 - mounted onto a virtual reality headset or embedded in PCs
 - Two infrared cameras and three infrared LEDs are used to detect hand and finger motions
- Project Soli
 - Track motion on a microscopic scale
 - Soli chip uses radar
 - It works by emitting electromagnetic waves with objects within the beam reflecting information back to the antenna



Leap Motion controller



[Project Soli](#)



[Motion Sense
on Google Pixel 4](#)

Gestural Input

- Gestures map movements to commands
 - Immediate and powerful interaction - almost anything one can do with one's hands could be a potential gesture
- problems?
 - Discoverability (not self revealing, not graphical)
 - Memorability (gestures need to be known and remembered)
 - Fatigue (movements over a long period of time can be very exhausting)
 - Gesture recognition accuracy

Buxton Collection

- <https://www.microsoft.com/buxtoncollection>

Mouse



3M
Ergonomic Mouse



Adesso Inc.
2-in-1 Optical
Keypad Calculator
Mouse AKP-170



Alias|Wavefront
Rockin' Mouse



Alias|Wavefront
Pad Mouse



Apple Inc.
Macintosh Model
M0100



Apple Inc.
iMac Round Mouse



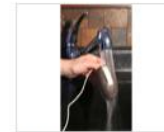
Apple Inc.
Macintosh ADB
Mouse



Apple Inc.
Magic Mouse



Appoint
MousePen



Belkin
Washable Mouse



Contour Design
Perfit Mouse Optical
Model PMO-M-L



Depraz
Swiss Mouse



Dimentor
Inspector 6DOF
Trackball Mouse



Freespace
Loop Pointer



Gyration
Air Mouse



Hanvon
Ink222 T&Mouse



HP
MoGo Slim



IBM
TrackPoint Mouse
G1



IBM
TrackPoint Mouse
G2



IBM
TrackPoint Mouse
G3

What did we cover

- Dimensions to Classify Computer Input
- Different keyboard designs
 - Qwerty and Dvorak Keyboard
 - “Virtual” Keyboards
 - Chording Keyboards
- Positional Input and Properties of Positional Input Devices
 - Absolute vs. Relative Positioning
 - Direct vs. Indirect Contact
 - Force vs. Displacement Sensing
 - Position vs. Rate Control
 - DOF (Dimensions) Sensed
- Gestural Input

Question?